To appear in Proceedings of Sinn und Bedeutung 30

Anaphora for Concepts, Kinds, and Parts in Dynamic Interpretation[1]

Manfred KRIFKA – *Leibniz-Zentrum Allgemeine Sprachwissenschaft (ZAS) Berlin*

Abstract. The article deals with anaphoric phenomena that involve concepts, kinds, specimens, parts and totalities. It sketches a theory that explains the use of singular vs. plural kind anaphors that assumes different morphosyntactic features for mass vs. count nouns. It discusses the phenomenon that concept antecedents are not restricted by anaphoric islands, and proposes that this is because they are name-like. It proposes an implementation within DRT and in compositional dynamic interpretation.

Keywords Anaphora, Concepts, Genericity, Partitives, DRT, Dynamic semantics, Mass/Count

1 Introduction

The prototypical type of anaphora consists in the explicit introduction of a discourse referent (dref) with an indefinite DP or a name, where this dref is anchored to an entity, followed by the uptake of this dref with a pronoun (Karttunen 1969). Consider (1), where *John* and *a spider* introduce and *he* and *it* pick up drefs, agreeing with their antecedent in gender and number.

(1) ***John*** *killed* ***a spider*** *because* ***he*** *was afraid of* ***it****.*

The drefs in (1) are anchored to entities. There are also anaphora that refer to kinds: In (2a), the kind-referring antecedent is a bare plural, in (2b), a definite singular, in (2c), a name, and in (2d), an indefinite DP in the so-called "taxonomic" reading.

(2) a. ***Spiders*** *are air-breathing arthropods.* ***They*** *are the largest order of arachnids.*
b. ***The tarantula*** *is a large spider.* ***It*** *looks threatening but is quite harmless.*
c. ***Lycosa tarantula*** *occurs in Southern Europe. People keep* ***it*** *sometimes as a pet.*
d. *The island was invaded by* ***a dangerous spider****. The government exterminated* ***it****.*

But kind anaphora may also have object-referring antecedents, as Krifka et al. (1995) point out with examples (3a,b). Notice the number mismatch between the singular antecedent and the plural anaphor in (3a); notice that (2b,c,d) also allow for plural anaphors.

(3) a. *John killed* ***a spider*** *because* ***they*** *are ugly.*
b. *John drank* ***some milk*** *even though he is allergic to* ***it****.*

Krifka et al. also mention anaphoric DPs like *one* and *some* that are anaphorically linked to indefinite DPs can introduce new drefs as in (4a,b). They appear to make use of the nominal concept introduced by the antecedent, e.g. the DP *one* can be spelled out as *one spider.*

[1] I would like to thank the anonymous reviewers and the audiences of the workshop *Mismatches in Anaphoric Relations* at the University of Graz, 29.-30. 5. 2025, and the *Sinn und Bedeutung* 30 at Goethe-University Frankfurt, 24.-26. 9. 2025. I also thank Peter Sutton and Luke Adamson for discussion.

(4) a. *John saw* ***a spider*** *and Mary saw* ***one*** *too.*
b. *John drank* ***some milk*** *and Mary drank* ***some*** *too.*

Krifka et al. furthermore mention that kind anaphora and *one*-anaphora are not restricted in the same way as those for objects. For example, they can be accessed even if introduced under the scope of negation, cf. (5a,b), which would block the uptake of drefs for objects, cf. (5c).

(5) a. *John <u>doesn't</u> own a* ***dog****. He is afraid of* ***them****. But Mary owns* ***one****.*
b. *Mary <u>didn't</u> buy any* ***milk****. She cannot digest* ***it****. But John bought* ***some****.*
c. *John <u>doesn't</u> own a* ***dog****.* ****It*** *is friendly.*

So far, we have discussed anaphora that refer to kinds (such as *them*) or to specimens of kinds (such as *one*). The same expressions can be used for partitive relationships. In (6a), *one* refers to a part of an entity, to be spelled out as 'one of the apples Mary bought', whereas in (6b), *them* refers to a totality of entities, to be spelled out as 'the teacups (that John owns)'.

(6) a. *Mary bought* ***some apples****. John ate* ***one****.*
b. *John found* ***a teacup*** *in the living room. He usually keeps* ***them*** *in the kitchen.*

While there exists work on such anaphora (cf. Webber 1979, Dahl 1984, Gernsbacher 1991, Lødrup 2012, Recasens et al. 2016, Weeber 2023), they have not been systematically integrated into frameworks of dynamic interpretation (but see LuperFoy 1991). I will proceed in three steps. In Section 2, I discuss the relevance of the count/mass distinction for the choice of grammatical number of anaphors. In Section 3, I propose an analysis within Discourse Representation Theory (DRT), which brings out the essence of the analysis in a particularly clear way. In Section 4 I will show how this analysis can be expressed in a framework of direct dynamic interpretation. This allows for a compositional semantic interpretation and reveals a mechanism for modelling the accessibility of antecedent DPs in the scope of operators like negation, as in (5). This mechanism shows that concepts and kinds are treated like names.

2 Kind Reference and the Mass / Count distinction

Examples (3a,b) have shown that antecedents that refer to objects allow for kind anaphora. We observe that, regardless of their grammatical number, count noun antecedents require plural anaphora as in (7a), whereas mass nouns require singular anaphora as in (7b). This even holds in cases in which the two constructions refer to the same individuals as in (8a,b,c):

(7) a. *John noticed* ***a spider / spiders / two spiders*** *in the bathroom.*
He has a phobia against ***them*** */ *it.*
b. *John noticed* ***mold / a spot of mold / two spots of mold*** *in the bathroom.*
He is allergic against ***it*** */ *them.*

(8) a. *There is a lot of* ***pollen*** *in the air. I am allergic against* ***it*** */ *them.*
b. *There are a lot of* ***pollen grains*** *in the air. I am allergic against* ***them*** */ ??it.*
c. *I discovered* ***a ragweed pollen grain*** *under the microscope. I am allergic against* ***them*** */ ??it.*

This problem is not peculiar to kind anaphora, but can be reproduced for entity anaphora:

(9) a. ***The pollen*** *we collected* ***is*** */ *are in the test tube. We will analyze* ***it*** */ *them.*
b. ***The pollen grains*** *we collected* ***are*** */ *is in the test tube. We will analyze* ***them*** */ *it.*

There are two ways to deal with this variation. The first is to assume that there is an ontological difference between count noun denotations and mass noun denotations. That is, *mold* and *spider* apply to entities of different sort, and this would also hold for *pollen* and *pollen grain*. The second way is to assume that there is just a formal difference: *mold* and *pollen* are mass nouns and are picked up by *it*, whereas *spider* and *pollen grain* are count nouns, and hence are picked up by *they/them*. The ontological difference may be not well-motivated in view of doublets like *pollen* and *pollen grain*, cf. Bale (2021) for further discussion. There is a proposal to introduce a level in-between morphosyntactic features and strictly truth-conditional properties, namely Type Theory with Records (Cooper 2012, and for mass/count nouns, Sutton & Filip 2016); this could be also used to determine the type of anaphoric uptake.

In the present article, I follow the second view and assume a morpho-syntactic distinction (see Krifka, to appear, for a worked-out version following the view that assumes an ontological distinction). That is, I treat the mass/count distinction as a formal difference that determines the morphological form of anaphoric expressions, similar to grammatical gender, as illustrated for German in (10), using the synonyms *Briefmarke* (feminine) and *Postwertzeichen* (neuter).

(10) a. *Peter klebte* ***die Briefmarke*** *auf den Umschlag.* ***Sie*** */ *Es fiel aber wieder ab.*
b. *Peter klebte* ***das Postwertzeichen*** *auf den Umschlag. *Sie /* ***Es*** *fiel aber wieder ab.*
'Peter glued the postage stamp on the envelope. But it fell off again.'

The restriction expressed by feminine *sie* and neuter *es* does not rely on an ontological distinction, as with *he, she* and *it* in English, which roughly refer to males, females and non-humans, respectively. Rather, it has to be expressed with respect to the formal gender feature of *Briefmarke* and *Postwertzeichen*, as indicated in (11a,b). In a similar way, the choice of kind-referring pronouns depends on whether the antecedent is a mass noun (resulting in *it*) or a count noun (resulting in *they*), as indicated in (12). This cannot be reduced to number agreement, as shown by in (3a) and (8c). The choice reflects how the pronouns would be spelled out as a kind-referring term, namely as *pollen* or *pollen grains.*

(11) a. … *die Briefmarke* [FEM] … *sie* [FEM] …
b. … *das Postwertzeichen* [FEM] … *es* [NEUT] …

(12) a. … *a lot of pollen* [MASS] … *it* [MASS] …
b. … *a lot of pollen grains* [COUNT] … *they* [COUNT] …
c. … *a pollen grain* [COUNT] … *they* [COUNT] …

Reference to kinds will be modelled following Chierchia (1998) by individual concepts (functions from indices to individuals, type **se**) that are derived from properties (functions from individuals to functions from individuals to truth values, type **set**). For this, I assume model frames that contain a set of indices **I** and a set of entities **U** that forms a join semi-lattice with join operation ⊔ and part relation ⊑. Chierchia (1998) defines a down operator "$^{\cap}$" that takes a property P and maps it to a function from indices i to the maximal entity in the extension of P at i, P(i). Maximality is defined by the sigma operator (Link 1983 – note that Chierchia uses iota ι for this). See (13a,b) for these definitions.

(13) a. $\sigma S = \iota x[S(x) \wedge \forall x'[S(x') \rightarrow x' \sqsubseteq x]]$, the unique maximal entity that S applies to
b. $^{\cap}P = \lambda i[\sigma[P(i)]]$ – Chierchia's down operator, the kind corresponding to property P

A sufficient condition that the requirement of the σ operator is satisfied is that S is not empty, $\exists x[S(x)]$, and that S is cumulative, that is, $\forall x[[S(x) \wedge S(y)] \rightarrow S(x \sqcup y)]$, or, in the general case, $\forall S'[\forall x[S'(x) \rightarrow S(x)] \rightarrow S(\sqcup S')]$. Mass nouns like *mold* and plural count nouns like *spiders* arguably have this cumulative interpretation. Singular count nouns like *spider* don't have it in general, as they apply to atoms. So, if x and y are different individuals, and it holds that $[\![spider]\!](i)(x)$ and $[\![spider]\!](i)(y)$, then $[\![spider]\!](i)(x \sqcup y)$ is not guaranteed to be true. In fact, it is true only if x and y happen to pick out the same individual.

We assume that the denotation of the plural noun *spiders* is derived by the plural operator, which denotes the closure of a predicate under the sum operation. This is often denoted by * (cf. Link 1983); to stress its nature as a closure under the join operation $\sqcup$, I write it here as superscript $^{\sqcup}$. It is defined in (14). Here, the first line states that S′ corresponds to a superset of S, and that S′ is cumulative; the second line states that S′ is the smallest set with this property.

(14) $^{\sqcup}S = \iota S'[\forall x[S(x) \rightarrow S'(x)] \wedge \forall x,y[S'(x) \wedge S'(y) \rightarrow S'(x \sqcup y)]$
$\wedge \forall S''[[\forall x[S(x) \rightarrow S''(x)] \wedge \forall x,y[S''(x) \wedge S''(y) \rightarrow S''(x \sqcup y)]] \rightarrow \forall x[S''(x) \rightarrow S'(x)]]$

The closure operator $^{\sqcup}$ is the semantic interpretation of plurality, as illustrated in (15.a). The plural NP *spiders* differs from *spider* insofar as its extensions are closed under the join operation. As a consequence, the down operator can be applied to it, cf. (15.b). When the down operator is applied to a predicate P like $[\![spider]\!]$ that is not cumulative but quantized, i.e. for which it holds that $\forall i \forall x[P(i)(x) \wedge P(i)(y) \wedge P(i)(x \sqcup y) \rightarrow x=y]$, then the down operator leads to an intension that is only defined if the extension of P is a singleton, cf. (15.b), as with *the Pope*.

(15) a. $[\![spiders]\!] = PL([\![spider]\!]) = \lambda P \lambda i[^{\sqcup}P(i)]([\![spider]\!]) = \lambda i^{\sqcup}[\![spider]\!](i)$
b. $^{\cap}[\![spiders]\!] = {}^{\cap}\lambda i^{\sqcup}[\![spider]\!](i) = \lambda i[\sigma[^{\sqcup}[\![spider]\!](i)]]$
c. $^{\cap}[\![spider]\!]$ defined only for i such that $\#([\![spider]\!](i)) = 1$

Mass nouns come from the lexicon as properties with cumulative extensions, hence they do not require a closure operation for the down operator, cf. (16). In fact, if S is cumulative, then $^{\sqcup}S = S$, that is, closure under join does not lead to any change. For this reason, we also have an absorption rule, $^{\sqcup\sqcup}S = {}^{\sqcup}S$.

(16) $^{\cap}[\![mold]\!] = \lambda i[\sigma[^{\sqcup}[\![mold]\!](i)]]$

In Section 3, I will develop a proposal for kind anaphora that take up discourse referents anchored to concepts. I assume that concepts are introduced by the NP of the antecedent DP, and that they form antecedents to kind anaphors. The plural kind anaphors *it* for mass nouns and *they* for count nouns pick up an antecedent property P and deliver a kind individual via the down operator $^{\cap}$, where *they* also introduces pluralization by the closure operator $^{\sqcup}$ as illustrated in (17a,b). As argued above, we implement the distinction that *they* picks up count nouns, and *it* picks up mass nouns, by a morphosyntactic feature. This is similar to grammatical gender; for example, the feminine singular pronoun *sie* would be analyzed as $\lambda P[\text{FEM}]$ $\lambda i \iota x[P(i)(x)]$.

(17) a. $\llbracket it \rrbracket = \lambda P[\text{MASS}]\ \lambda i\, {}^{\cap}P(i)$
b. $\llbracket they \rrbracket = \lambda P[\text{COUNT}]\ \lambda i\, {}^{\cap\sqcup}P(i)]$

The examples in (18) illustrate four cases in which the NP contained in a DP introduces a concept that is then picked up by a kind anapher. The selection of *it* vs. *them* is determined by the morphosyntactic features [MASS] and [COUNT]. While the kind pronoun *it* just expresses the down operator ${}^{\cap}$, *they* also applies the plural-specific closure under join ${}^{\sqcup}$. This allows for the number mismatch in the case of (18c). In the case of (18d), the plural pronoun can be applied as well, as one of the join closures is cancelled due to the absorption rule (16). Note that in this case the singular pronoun *it* could have been used as well, were it not for the feature mismatch of a [COUNT] concept and a [MASS] pronoun.

(18)	DP antecedent	Kind anapher	Interpretation
a.	[DP *some* **[NP *mold*][MASS]**]	$\llbracket it \rrbracket(\llbracket mold \rrbracket)$	$\lambda i\, {}^{\cap}\llbracket mold \rrbracket(i)$
b.	[DP *two spots of* **[NP *mold*][MASS]**]	$\llbracket it \rrbracket(\llbracket mold \rrbracket)$	$\lambda i\, {}^{\cap}\llbracket mold \rrbracket(i)$
c.	[DP *a* **[NP *spider*][COUNT]**]	$\llbracket they \rrbracket(\llbracket spider \rrbracket)$	$\lambda i\, {}^{\cap\sqcup}\llbracket spider \rrbracket(i)$
d.	[DP *two* **[NP *spiders*][COUNT]**]	$\llbracket they \rrbracket(\llbracket spiders \rrbracket)$	$\lambda i\, {}^{\cap\sqcup}\llbracket \text{spiders} \rrbracket(i)$
			$= \lambda i\, {}^{\cap\sqcup\sqcup}\llbracket spider \rrbracket(i)$
			$= \lambda i\, {}^{\cap\sqcup}\llbracket spider \rrbracket(i)$

This is the way how kind-referring expressions can be construed from concepts. But we have seen in (2c-d) that kinds can also be picked up by singular pronouns. Hence, the reference objects of singular definite generics, named generics, and kinds in taxonomies also occur as atoms in **U**. For example, the individual concept $\lambda i\, {}^{\cap\sqcup}\lambda x[\llbracket tarantula \rrbracket(i)]$ is also an abstract object in **U**. As such, it is the referent of the name *the tarantula*. It is not in the extension of the regular predicate $\llbracket spider \rrbracket$, which only applies to concrete spiders, but of the taxonomic interpretation, $\text{TX}(\llbracket spider \rrbracket)$, which applies to the subkinds of the kind *spider*.

This is the way how kind-referring expressions can be constructed from concepts, or nominal intensions. We have seen in examples (2c-d) that kinds can also be picked up by singular pronouns, even in the case of count nouns. I assume that the reference object of singular definite generics, named generics, and kinds in taxonomies are treated as atoms in **O**, hence the singular anaphoric reference. Singular kind reference is possible only with a few "established" kinds, in contrast to the kinds that can be constructed by the down operator ${}^{\cap}$ from concepts in general.

This concludes our discussion of the mass/count distinction and its relevance for the interpretation of kind anaphora. We now turn to a modelling within Discourse Representation Theory.

3 Concept, Kind and Partitive Anaphora in DRT

Let us assume a standard DRT framework in which texts of syntactically structured sentences with reference indices are put through an algorithm that constructs DRSs (cf. Kamp & Reyle 1994). I will differentiate between kind anaphora and concept anaphora. In examples like (5a,b), the pronouns *them* and *it* refer to kinds, whereas the DPs *one* and *some* pick up the concepts 'dog' and 'milk'; they can be spelled out by *one dog*, or *some milk*.

Evidence for the distinction between concept anaphora and kind anaphora comes from (19). The second sentence of (19a) can be read as 'John is afraid of dogs', presumably because *be afraid of* allows for kind objects but 'dogs from the animal shelter downtown' does not constitute a kind (cf. Carlson 1977 for bare nouns like *people in the next room*). But the second sentence of (19b) has as a preferred reading 'Mary got a dog from the animal shelter downtown'. This would not be possible if the anaphoric relationship were mediated by kind anaphora

(19) a. *John didn't get a* ***dog*** *from the animal shelter downtown. He is afraid of* ***them****.*
b. *John didn't get a* ***dog*** *from the animal shelter downtown. But Mary got* ***one****.*

To start with, consider example (20). (20a) introduces a dref x_1 for John, a dref x_2 for the nominal property *dog*, a dref x_3 that is anchored to an entity that falls under this property at the current world/time index i_0, and the condition that at this index, John owns that dog. For reasons discussed in Section (3), I assume that x_4 is marked as "count", by [C]. As indices are essential for the reconstruction of kinds, I also assume a dref i_0 as the index of the situation. I assume that names like *John* are index-independent, in contrast to nominal predicates like *dog*.

(20) a. *John*$_1$ *owns* [$_{DP}$ *a*$_3$ [$_{NP}$ *dog*]$_2$]. c. *Mary*$_4$ *owns* [$_{DP}$ *an*$_5$ [$_{NP}$ *aggressive* [$_{NP}$ *one*]$_3$].
b. *It*$_3$ *is very friendly.* c′. *Mary*$_4$ *owns* [$_{DP}$ *two*$_5$ [$_{NP}$ *aggressive* [$_{NP}$ *ones*]$_3$].

$i_0\ x_1\ x_{2[C]}\ x_3$	$x_4\ x_5$	$x_4\ x_4$
x_1=John $x_{2[C]}=\lambda i\lambda x[\text{dog}(i)(x)]$ $x_2(i_0)(x_3)$ $\text{own}(i_0)(x_3)(x_1)$ $\text{friendly}(i_0)(x_3)$	x_4=Mary $x_{2[C]}(i_0)(x_5)$ $\#(x_5)=1$ $\text{aggressive}(i_0)(x_5)$ $\text{own}(i_0)(x_5)(x_4)$	x_4=Mary $^{\sqcup}x_{2[C]}(i_0)(x_5)$ $\#(x_5)=2$ $\text{aggressive}(i_0)(x_5)$ $\text{own}(i_0)(x_5)(x_4)$

In (20b), the pronoun *it* takes up the dref x_3 for the dog that John owns and introduces the condition that its anchor is friendly at i_0. (20.c) introduces a dref x_4 for Mary; it also picks up the index x_2 that is anchored to the concept 'dog' and introduces a dref x_5 that is anchored to an entity that falls under this property. The singular morphology of the anaphor *one* leads to the condition that the cardinality of x_5 is 1, i.e. it is an atom in the universe **U**. It is also stated that the anchor of x_5 is aggressive at i_0 and that Mary owns it at i_0. In the variant (20.c′) the plural anaphora *ones* picks up the property x_2 and introduces a dref x_5 that falls under the $\sqcup$-closure of this property at i_0. The numeral *two* requires that x_5 is anchored to an element of that consists of two atoms. The NP-anaphor *one* selects for count nouns; the corresponding anaphor for mass nouns in this case is null, as in *John bought blue paint and Mary green.*

Example (20a) illustrated the use of *one* as a concept anaphor. But *one* can be used in cases as in (4a,b), where it contrasts with *some* for mass nouns. I propose that in this case, there is a null concept anaphor, and *one* is in the determiner position. Consider example (21a). In (21b), the kind anaphor *them* picks up the dref x_2 referring to the count concept 'dog', and introduces the corresponding kind individual, with the interpretation illustrated in (18c). The double role of *them*, as picking up the dref associated to the concept antecedent, [$_{NP}$ *dog*]$_2$, and introducing an object dref x_4 that falls under this concept, is indicated by the double index on the pronoun. The third sentence (21c) picks up the dref x_3 that is anchored to the dog that John owns.

(21) a. *John*$_1$ *owns* [$_{DP}$ *a*$_3$ [$_{NP}$ *dog*]$_2$].
b. *He*$_1$ *likes them*$_{2,4}$.
c. *It*$_3$ *is very friendly.*
d. *Mary*$_5$ *owns* [$_{DP}$ *one*$_6$ [$_{NP}$]$_2$] (*too*). / d′. *Mary*$_5$ *owns* [$_{DP}$ *two*$_6$ [$_{NP}$]$_2$].

$i_0\ x_1\ x_{2[C]}\ x_3$	x_4		$x_5\ x_6$	
x_1=John	$x_4=\lambda i^{\cap\sqcup}x_{2[C]}(i)$	friendly(i_0)(x_3)	x_5=Mary	
$x_{2[C]}=\lambda i\lambda x[\text{dog}(i)(x)]$	like(i_0)(x_4)(x_1)		${}^{\sqcup}x_2(i)(x_6)$	
$x_2(i_0)(x_3)$			$\#(x_6)=1$	/ $\#(x_6)=2$
own(i_0)(x_3)(x_1)			own(i_0)(x_6)(x_5)	

Sentences (21d) is similar to (20c′) as it picks up the concept anaphora x_2. However, now the concept anaphor is the empty NP, and *one* is rather a numeral in the determiner position: The plural version (21d′) is *Mary owns two*, not **Mary owns two ones.* The NP anaphor *one(s)* is required only in case the NP is modified by an adjective as in (20c,c′) (in contrast to German, where it is always empty; e.g. (20c) is translated as *Mary hat einen agressiven*). The two distinct roles of *one* are particularly evident in *John owns two friendly dogs and* [$_{DP}$ *one* [$_{NP}$ *aggressive* [$_{NP}$ *one*]]], where the first *one* fills the determiner position, and the second, the NP position (cf. Jackendoff 1977). The null NP anapher is unspecific for the count/mass feature, as it also occurs with mass noun, as in *Mary drank wine, and John drank* [$_{DP}$ *some* [$_{NP}$]] *too*.

The concept anaphors [$_{NP}$] and [$_{NP}$ *one(s)*] have to be contrasted with partitive anaphora, illustrated in (22). The indefinite antecedent introduces three drefs: x_3 for the concept ‘apples’, x_4 for the sum of all (relevant) apples in the context, and x_5 for one of these apples. The partitive anaphor *two* picks up x_4 and introduces a part of it, x_8, that consists of two atoms. In contrast to concept anaphors, the head of the partitive construction is a PP, not an NP, as it has to be spelled out as *two **of** the apples.* The kind anaphor *them* picks up the concept anaphor x_3 and introduces a dref x_9 for the kind, as above. As the first two sentences are in the past tense, their world/time indices are shifted to the past.

(22) a. *John*$_2$ *took*$_1$ [$_{DP}$ *one*$_5$ [$_{PP}$ *of* [*the* [$_{NP}$ *apples*]$_3$]$_4$]]
b. *and Mary*$_7$ *took*$_6$ [$_{DP}$ *two*$_8$ [$_{PP}$]$_4$].
c. *She*$_7$ *likes them*$_{3,8}$.

$i_0\ i_1\ x_2\ x_{3[C]}\ x_4\ x_5$	$i_6\ x_7\ x_8$	x_9
$i_1<i_0$	$i_6<i_0$	
x_2=John	x_7=Mary	$x_9=\lambda i^{\cap\sqcup}x_{3[C]}(i)$
$x_{3[C]}=\lambda i\ {}^{\sqcup}\text{apple}(i)$	$x_8 \sqsubseteq x_4$	like(i_0)(x_9)(x_7)
$x_4=\sigma\, x_{3[C]}(i_0)$	$\#(x_8)=2$	
$x_5 \sqsubseteq x_4$	take(i_6)(x_8)(x_7)	
$\#(x_5)=1$		
take(i_1)(x_5)(x_2)		

Similar cases with mass nouns are illustrated with example (23).

(23) a. *John*$_2$ *bought*$_1$ [$_{DP}$ *two*$_4$ *kilogram* [$_{pp}$ *of* [$_{NP}$ *honey*]$_3$]].
b. *He*$_2$ *likes it*$_{3,5}$.
c. *He*$_2$ *put*$_6$ *it*$_4$ *into the pantry.*
d. *Mary*$_8$ *bought*$_7$ [$_{DP}$ *some*$_9$ [$_{NP}$]$_3$] (*too*).

$i_0\ i_1\ x_2\ x_{3[M]}\ x_4$	x_5	i_6	$i_7\ x_8\ x_9$
$i_1<i_0$		$i_6<i_0$	$i_7<i_0$
x_2=John	$x_5=\lambda i\sqcup x_{3[M]}(i)$	put-in-pantry	x_8=Mary
$x_{3[M]}=\lambda x[\text{honey}(i)(x)]$	like(i_0)(x_5)(x_2)	(i_6)(x_4)(x_2)	$x_{3[M]}(i_7)(x_9)$
$x_4=x_3(i_1)$			buy(i_7)(x_9)(x_8)
kg($\boldsymbol{\varepsilon}$(x_4))=2			
buy(i_1)(x_4)(x2)			

Just as *one*-DPs can refer to a specimen of a kind or a part of a totality, the pronoun *them* can refer to a kind or a totality. In (24), *them* is to be spelled out as *the teacups*, hence it makes use of a pluralized concept. The operator ι refers to the maximum of entities that fall under ‘teacups’, restricted by the context (not indicated here).

(24) A: *There is* [$_{DP}$ *a*$_2$ [*teacup*]$_1$] *in the living room*. B: *I keep* [*them*] *in the kitchen*.

i_0 $x_{1[C]}$ x_2	x_3
$x_{1[C]}$=λiλx[teacup(i)(x)] $x_{1[C]}(i_0)(x_2)$ in_living_room(i_0)(x_2)	$x_3=\sigma^{\sqcup}x_{1[C]}(i_0)$ keep_in_kitchen(i_0)(x_3)(B)

We now turn to the important property of concept drefs, that they do not observe anaphoric island restrictions but are globally available, even if introduced in the scope of operators like a negation, cf. (5a,b). This means that they are introduced into the global DRS, not into a local one that is created by operators like negation, disjunction, conditionalization, quantization, or modal operators. This is illustrated with negation with example (25).

(25) a. *John*$_1$ *doesn’t own* [$_{DP}$ *a*$_3$ [$_{NP}$ *dog*]$_2$].
b. *He*$_1$ *is afraid of them*$_{2,4}$.
c. *Mary*$_5$ *owns* [$_{DP}$ *two*$_6$ [$_{NP}$]$_2$].

i_0 x_1 x_2	x_4	x_5 x_6
x_1=John $x_{2[C]}$=λiλx[dog(i)(x)] ¬ [x_3 | $x_2(i_0)(x_3)$, own(i_0)(x_3)(x_1)]	$x_4=\lambda i\ {}^{\cap\sqcup}x_{2[C]}(i)$ afraid-of(i_0)(x_4)(x_1)	x_5=Mary $^{\sqcup}x_2(i_0)(x_6)$ #(x_6)=2 own(i_0)(x_6)(x_5)

Note that the concept dref x_2 is introduced in the main box, which makes it possible to be picked up in subsequent sentences. The dref x_3 for the DP *a dog* is introduced in the local DRS, in the scope of the negation operator, hence it could not be taken up in subsequent sentences, and continuations like *it is very friendly* are ruled out.

Let us consider an example involving a generic predication. In (26a), the predication itself is not about the kind, but the head of the indefinite DP introduces a kind dref that is introduced in the global box, as in (25). The generic predication introduces a quantificational structure, with GEN as the generic operator (cf. Krifka et al. 1995). (26b,b′) illustrates two continuations.

(26) a. [$_{DP}$ *a* [$_{NP}$ *dog*]$_1$]$_2$ *is a good companion*.

i_0 x_1		
$x_{1[C]}$=λiλx[dog(i)(x)]		
[i_1 x_2 | $x_{1[C]}(i_1)(x_2)$, #(x_2)=1]	GEN	[good_companion(i_1)(x_2)]

b. *They*$_{1,3}$ *are friendly to children.* b′. *It*$_2$ *is friendly to children.*

x_3
$x_3=\lambda i^{\cap\sqcup}x_{1[C]}(i_1)$

$i_1\ x_4$ $x_4 \sqsubseteq x_3(i_1)$	GEN	friendly$(i_1)(x_4)$

$i_1\ x_2$ $x_{1[C]}(i_1)(x_2)$ $\#(x_2)=1$ good_companion$(i_1)(x_2)$	GEN	friendly$(i_1)(x_2)$

In (26b), *they* picks up the concept dref x_1 and introduces a dref x_3 for the kind that corresponds to that concept. The sentence expresses a generic quantification itself that ranges over specimens x_4 of that kind. The analysis is simplified, as it omits reference to children and to situations. In contrast, (26b′) is an instance of modal subordination (cf. Roberts 1989), which requires the construction of a union of the two DRSs of the first clause (26a) that forms the antecedent DRS of a generic interpretation. The two sentences have the same meaning.

The following example contrasts *one*-anaphora that involve partitive readings with *one*-anaphora that refer to specimens of a concept. Sentence (27a) can be continued by (27b) or (27b′).

(27) a. *Mary*$_2$ *bought*$_1$ [$_{DP}$ [$_{Det}$]$_4$ [$_{NP}$ *apples*]$_3$].
b. *John*$_6$ *ate*$_5$ [$_{DP}$ *one*$_7$ [$_{PP}$]$_4$]. / b′. *Sue*$_8$ *bought*$_7$ [$_{DP}$ *one*$_9$ [$_{NP}$]$_2$] *too.*

$i_0\ i_1\ x_2\ x_3\ x_4$ $i_1<i_0$ x_2=Mary $x_{3[C]}=\lambda i^{\sqcup}\lambda x[\text{apple}(i)(x)])$ $x_{3[C]}(i_1)(x_4)$ buy$(i_1)(x_4)(x_2)$	$i_5\ x_6\ x_7$ $i_5<i_0$ x_6=John $x_7 \sqsubseteq x_4$ $\#(x_5)=1$ eat$(i_5)(x_7)(x_6)$	$i_8\ x_9\ x_{10}$ $i_8<i_0$ x_9=Sue $^{\sqcup}x_{3[C]}(i_8)(x_{10})$ $\#(x_{10})=1$ buy$(i_2)(x_{10})(x_9)$

In the partitive anaphora case (27b), [$_{PP}$] picks up an object dref, x_4, and the DP introduces a dref x_7 that is part of that object. In the kind anaphora case (27b′), [$_{NP}$] picks up a concept dref, x_3, and the DP introduces an entity or entities x_{10} of that concept at the evaluation index. One requirement of this is that the empty head NP applies to sum objects; hence *Mary bought an apple. John ate one* does not have this reading, which would be expressed by *John ate it.* Notice that there are partitive constructions and specimen constructions with mass nouns as well, as in *Mary bought honey. John ate some / John bought some too.*

I conclude with a case in which an indefinite antecedent introduces a kind, as in the taxonomic interpretation in (2c). The function TX yields taxonomic readings, i.e. TX(⟦*beetle*⟧)(i) applies to subkinds of beetles at i, which are elements of **A**. The dref x_3 is anchored to the concept that applies to subkinds of beetles, and the dref x_4, to dangerous subkinds. The dref x_5 is such a subkind, and the first sentence states that the island Efate (x_2) is invaded by it. In the second sentence, the pronoun *it* refers to that subkind via the dref x_5. In the third, *one*-anaphora introduces a new subkind x_{10} via x_4 or alternatively x_3, the concepts of (dangerous) subkinds of beetles. The pronoun *one* can also refer to exemplars as in (28d), in which case they have to be spelled out as *one of them*. This shows that the *of*-construction allows, in addition to the partitive reading, a reading that refers to specimens of a kind.

(28) a. [*Efate*]$_2$ *was*$_1$ *invaded by* [$_{\text{DP}}$ *a*$_5$ [$_{\text{NP}}$ *dangerous* [$_{\text{NP}}$ *beetle*]$_3$]$_4$]
b. *It*$_5$ *was*$_6$ *exterminated.*
c. [*Malakula*]$_7$ *was*$_8$ *invaded by* [$_{\text{DP}}$ *one*$_9$ [$_{\text{NP}}$]$_4$] *too.* / c′. *John*$_{10}$ *saw*$_{11}$ [$_{\text{DP}}$ *one*$_{12}$ [$_{\text{NP}}$]$_5$]

$i_0\ i_1\ x_2\ x_3\ x_4$	i_6	$i_8\ x_7\ x_9$	$i_{11}\ x_{10}\ x_{12}$
$i_1 < i_0$ x_2=Efate $x_{3[C]} = \lambda i[\text{K(beetle)}(i)]$ $x_{4[C]} = \lambda i \lambda x[\text{K(beetle)}(i)(x)$ $\wedge\ \text{dangerous}(i)(x)]$ $x_{4[C]}(i_0)(x_5)$, $\#(x_5)=1$ $\text{invaded\_by}(i_1)(x_2)(x_5)$	$i_6 < i_0$ exterminated $(i_6)(x_5)$	$i_8 < i_0$ x_7=Malakula $^{\cup}x_{4[C]}(i_8)(x_9)$ $\#(x_9)=1$ invaded_by $(i_8)(x_7)(x_9)$	$i_{11} < i_0$ x_{10}=John $^{\cap\cup}x_{5[C]}(i_{11})(x_{13})$ $\#(x_{13})=1$ $\text{see}(i_{11})(x_{13})(x_{12})$

In [illegible] d [illegible] t [illegible] ed to intensional entities, specifically, properties. This has bee proposed for other phenomena as well, such as propositional anaphora and intensional drefs (Snider 2017, Hofmann 2024).

4 Concept, Kind and Partitive Anaphora in Dynamic Semantics

In Section 3, I outlined my proposal concerning anaphoric reference to concepts and kinds in the representational framework of DRT. In this section, I will develop an account in terms of dynamic interpretation, which takes a representation of the current context and a syntactic structure and maps them directly to semantic objects, in a compositional way.

Many accounts of direct dynamic interpretation have been proposed, e.g. Heim (1983), Groenendik & Stokhof (1990), and others. The version I will develop here is most closely derived from Rooth (1987) and Krifka (1993). In this framework, syntactically structured and labelled expressions are interpreted as functions from input assignments to output assignments to intensions (functions from world-time indices to other semantic objects, e.g. truth values in the case of propositions). Assignments are partial functions from the set of natural numbers to semantic objects. The value of assignments is not restricted to individuals (entities or pluralities) but includes world-time indices and individual concepts. I use g, h, k for assignments,(partial functions from natural numbers to meanings). I write g_n for g(n), the value that g assigns to the number n. I write g<h if the assignment h extends the assignment g, that is, h restricted to the domain of g is identical to g. More specifically, I write $g<_{n\dots n'}h$ if g<h, and the domain of h differs from g insofar as it contains the numbers n…n′.

The basic format of interpretation (to be modified later) is illustrated in example (29a) for the sentence *John saw Mary.* It is a function from input assignments g to output assignments h to functions from world/time indices i to truth values. Let us take **a** for the type of assignments, the type of (29) can then be rendered as **aast**. More specifically, h extends g by the indices 1,2,3 and maps the dref 1 to an index temporally before i, the dref 2 to John, and the dref 3 to Mary, under the condition that the entities mapped by h(2) and h(3) stand in the see-relation at the index mapped by h(1). F is the function that assigns basic expressions their non-dynamic meaning; the meaning of an expression α dependent on an index i is given as F(α)(i). Names are treated as index-independent, for simplicity. In order to keep representations more compact and improve readability, I will use the format in (29b).

(29) a. ⟦PAST$_1$ [*John*$_2$ *see Mary*$_3$]⟧
$= \lambda g\lambda h\lambda i[g<_{1,2,3}h \wedge h_1<i \wedge h_2=F(\mathit{John}) \wedge h_3=F(\mathit{Mary}) \wedge F(\mathit{see})(h_1)(h_3)(h_2)]$
b. $\lambda ghi[g<_{123}h, h_1<i, h_2=F(\mathit{J}), h_3=F(\mathit{M}), F(\mathit{see})(h_1h_3h_2)]$

Such meanings update information states, or contexts, of the participants of a conversation. We model contexts as sets of pairs ⟨i,g⟩ of world/time indices i and assignments g (cf. Heim 1983). I assume that all assignments g of a context c have the same domain, which we call DOM(c). The update c + p of a context c with a dynamic sentence meaning p is accomplished by rule (30) as a pointwise update of all pairs ⟨i, g⟩ in c by p, as exemplified in (31).

(30) $c + p = \{\langle i,h\rangle \mid \exists g[\langle i,g\rangle \in c \wedge p(ghi)]\}$

(31) c_0 + ⟦PAST$_1$ [*John*$_2$ *see Mary*$_3$]⟧ =
$\{\langle i,h\rangle \mid \exists g[\langle i,g\rangle \in c_0, g<_{123}h, h_1<i, h_2=F(\mathit{J}), h_3=F(\mathit{M}), F(\mathit{see})(h_1h_3h_2)]\} = c_1$

Sentence (29) can be continued by sentences with anaphoric expressions that pick up the drefs that were introduced in (29). This is shown for *he waved* in (32), where *he* picks up the dref 2. The past operator is anaphoric, as it relates its index to the index of the past operator of the preceding sentence (interpreted by a relation "-" that includes overlap and succession). The resulting update function presupposes that the indices 1, 2, 3 are in the domain of the input g. F(*MALE*)(ix) expresses that at i, x is a male person.

(32) ⟦PAST$_{14}$ [*he*$_2$ *wave*]⟧
$= \lambda ghi[g<_4h, h_4<i, h_1\text{-}h_4, F(\text{MALE})(h_4h_2), F(\mathit{wave})(h_4h_2)]$

This sentence can be interpreted with respect to the context c_1 constructed in (31):

(33) c_1 + ⟦PAST$_{14}$ [*he*$_2$ *wave*]⟧
$= \{\langle i,h\rangle \mid \exists g[\langle i,g\rangle \in (31) \wedge (32)(ghi)]\}$
$= \{\langle i,h\rangle \mid \exists g[\langle i,g\rangle \in c \wedge g<_{1234}h, h_1<i, h_2=F(\mathit{J}), h_3=F(\mathit{M}), F(\mathit{see})(h_1h_3h_2),$
$h_4<i, h_1\text{-}h_4, F(\text{MALE})(h_4h_2), F(\mathit{wave})(h_4h_2)]$

However, this representation is problematic when we consider constructions like negation, which we interpret as in (34) as a sentential operator. Negation forces input assignment g and output assignment h to be the same, and impose the condition on the indices i that the input assignment cannot be extended so that the proposition p, a variable of type **aast**, is satisfied.

(34) ⟦NEG⟧ $= \lambda p\lambda ghi[g=h, \neg\exists k[g\leq k, p(gki)]]$

When we apply this operator to (29), the drefs for John and Mary become inaccessible for anaphoric uptake in subsequent sentences, as they are introduced in the scope of ¬∃k. We expect this for non-specific indefinite expressions, as in *John didn't see a woman*, but not for names. What we need is the equivalent for the construction rule in DRT that name drefs are introduced in the global DRS, not the local DRSs. According to Muskens (1996), drefs of names are related to external anchors, which results in treating names as constants. Here, I propose that names presuppose that their dref already exists in the input assignment. This leads to the following representation of the meaning of (29), an update function that presupposes that the input g contains the drefs 2 and 3 with the indicated interpretation.

(29′) $\lambda ghi{:}\ {\scriptstyle g_2=F(\mathit{J}),\ g_3=F(\mathit{M})}\ [g<_1h, h_1<i, F(\mathit{see})(h_1h_3h_2)]$

When we apply ⟦NEG⟧ to (29′), the drefs for the names project to the input assignment, and hence are available for update in subsequent sentences:

(35) ⟦NEG⟧(29′) $= \lambda ghi$: $g_2{=}F(J), g_3{=}F(M)$ $[g{=}h, \neg\exists k[g{<_1}k, k_1{<}i, F(\textit{see})(h_1h_3h_2)]]$

I use the notation λx: Presupp[x] […x…] for functions from x to […x…] that are restricted to those x for which Presupp[x] is true. That is, the function is not defined if Presupp[x] is false. This affects the definedness conditions for larger formulas, following Kleene's strong interpretation of three-valued logic. That is, whenever a formula contains a subformula that is undefined, the formula is undefined.

In case an input assignment does not contain a presupposed dref in its domain, it can be accommodated (cf. Lewis 1979). This accommodation just introduces a dref that is uniquely anchored to the bearer of the name but does not add any factual information. We amend the update rule for contexts (30) as follows:

(36) If c + p is undefined because DOM(c) does not fit the presuppositions of p,
then perform the update c′ + p instead, where $c' = \{\langle i,g'\rangle \mid \exists g[\langle i,g\rangle\in c \wedge g'$ is
the **unique minimal** extension of g such that p(g′) is defined$]\}$

This accommodation is uncontroversial, as it does not ask the addressee to assume any factual information. In this, it is different from cases like A: *My tarantula has escaped,* which may prompt B, not knowing that A is an arachnophile, to react with B: *Oh, you have a tarantula?* Possible rejections could be that the addressee does not know which person is meant because the name is not known to the addressee, or that it is ambiguous. Later, the same mechanism will be used for the accommodation of drefs for concepts, like 'dog', which are also unique, and which participants are expected to be familiar with due to their knowledge of language.

Meanings can be interpreted compositionally, as illustrated in (37) for *John saw Mary*. Here, the propositional variable p is of type **aast**, the variable P for subject meanings is of type **eaast**, and the object variable R is of type **eeaast**. Meaning composition is by functional composition.

(37) a. $[\![\text{PAST}_1]\!] = \lambda p\lambda ghi\exists k[g{<_1}h, h_1{<}i, p(hkh_1)]$
b. $[\![\textit{John}_2]\!] = \lambda P\lambda ghi$: $g_2{=}F(J)$ $[P(g_2)(ghi)]$
c. $[\![\textit{see}]\!] = \lambda y\lambda x\lambda ghi[g{=}h, F(\textit{see})(iyx)]$
d. $[\![\textit{Mary}_3]\!] = \lambda R\lambda x\lambda ghi$: $g_3{=}F(M)$ $[R(g_3)(x)(ghi)]$
e. $[\![\textit{see Mary}_3]\!] = [\![\textit{see}]\!]([\![\textit{Mary}_3]\!])$
$= \lambda x\lambda ghi$ $g_3{=}F(M)$ $[g{=}h, F(\textit{see})(ih_3x)]$
f. $[\![\textit{John}_2\ \textit{see Mary}_3]\!] = [\![\textit{see Mary}\ _3]\!]([\![\textit{John}_2]\!])$
$= \lambda ghi$: $g_2{=}F(J), g_3{=}F(M)$ $[g{=}h, F(\textit{see})(ih_3h_2)]$
g. $[\![\text{PAST}_1\ \textit{John}_2\ \textit{see Mary}_3]\!] = [\![\text{PAST}_1]\!]([\![\textit{John}_2\ \textit{see Mary}_3]\!])$
$= \lambda ghi$: $g_2{=}F(J), g_3{=}F(M)$ $[g{<_1}h, h_1{<}i, F(\textit{see})(h_1h_3h_2)]$

Subject DPs and object DPs have slightly different types, **(eaast)aast** and **(eeaast)eaast**. Such differences should be handled by the interpretation of case labels. (38) illustrates this with deriving the non-dynamic interpretation of the name *Mary* in (38a) via a labelling operation with the dref index 3 in (38c) and an interpretation of the accusative feature in (38e).

(38) a. $\llbracket \textit{Mary} \rrbracket = \text{F}(\textit{M})$, type **e**
b. $\llbracket 3 \rrbracket = \lambda y \lambda x \lambda ghi_{\,g_3=y}\ [g=h,\ x=g_3]$, type **eeaast**
c. $\llbracket [_{\text{DP}}\ \textit{Mary}_3] \rrbracket = \llbracket 3 \rrbracket(\llbracket \textit{Mary} \rrbracket),\ = \lambda x \lambda ghi_{\,g_3=\text{F}(M)}\ [g=h,\ x=g_3]$, type **eaast**
d. $\llbracket \text{ACC} \rrbracket = \lambda P \lambda R \lambda x \lambda ghi \exists k \exists y[g \leq k \wedge P(y)(gki),\ R(yx)(ghi)]$, **(eaast)(eeast)eaast**
e. $\llbracket [_{\text{DP}}\ \textit{Mary}_3]_{\text{ACC}} \rrbracket = \llbracket \text{ACC} \rrbracket(\llbracket [_{\text{DP}}\ \textit{Mary}_3] \rrbracket)$
$= \lambda R \lambda x \lambda ghi \exists k \exists y[g \leq k,\ \lambda x \lambda ghi_{\,g_3=\text{F}(M)}\ [g=h,\ x=g_3](y)(gki),\ R(yx)(khi)]$
$= \lambda R \lambda x \lambda ghi{:}\ _{g_3=\text{F}(M)}\ [R(g_3x)(ghi)]$, type **(eeast)eaast**

Meanings of type **aast** like (37) can update a context c, triggering accommodation to a minimal variant c′ following (36) when drefs 2 and 3 are not defined. Further updates with expressions like *He waved* are possible that pick up the drefs in the accommodated context. I represent this context update in the following way, where the presuppositions for the input assignment g that can be accommodated are conspicuous:

(39) $c_0 + (37) = \{\langle i,h \rangle \mid \exists g{:}\ _{g_2=\text{F}(J),\ g_3=\text{F}(M)}\ [\langle g,i \rangle \in c_0,\ [g <_1 h,\ h_1 < i,\ \text{F}(\textit{see})(h_1h_3h_2)]\}$

We now turn cases that introduce concept anaphora. We follow the DRT representation of examples like (25) and (26), and start with example (20a), which is derived in (40).

(40) a. $\llbracket \textit{John}_1 \rrbracket = \lambda P \lambda ghi_{\,g_1=\text{F}(J)}\ [P(g_1)(ghi)]$
b. $\llbracket \textit{own} \rrbracket = \lambda y \lambda x \lambda ghi[g=h,\ \text{F}(\textit{own})(iyx)]$
c. $\llbracket \textit{a}_3 \rrbracket = \lambda P \lambda R \lambda x \lambda ghi \exists k \exists k'[g <_3 k,\ P(k_3)(kk'i),\ R(k'_3x)(k'hi)]$
d. $\llbracket \textit{dog}_2 \rrbracket = \lambda x \lambda ghi_{\,g_2=\text{F}(\textit{dog}),\ \text{F}(\text{C})(g_2)}\ [g=h,\ g_2=x]$
e. $\llbracket [_{\text{DP}}\ \textit{a}_3\ [_{\text{NP}}\ \textit{dog}]_2] \rrbracket = \llbracket \textit{a}_3 \rrbracket(\llbracket \textit{dog}_2 \rrbracket)$
$= \lambda R \lambda x \lambda ghi_{\,g_2=\text{F}(\textit{dog}),\ \text{F}(\text{C})(g_2)}\ \exists k[g <_3 k,\ k_2(ik_3),\ R(k_3x)(khi)]$
f. $\llbracket \textit{own}\ [_{\text{DP}}\ \textit{a}_3\ [_{\text{NP}}\ \textit{dog}]_2] \rrbracket = \llbracket [_{\text{DP}}\ \textit{a}_3\ [_{\text{NP}}\ \textit{dog}]_2] \rrbracket(\llbracket \textit{own} \rrbracket)$
$= \lambda x \lambda ghi{:}\ _{g_2=\text{F}(\textit{dog})}\ [g <_3 h,\ h_2(ih_3),\ \text{F}(\textit{own})(ih_3x)]$
g. $\llbracket \textit{John}_1\ \textit{own}\ [_{\text{DP}}\ \textit{a}_3\ [_{\text{NP}}\ \textit{dog}]_2] \rrbracket = \llbracket \textit{own}\ [_{\text{DP}}\ \textit{a}_3\ [_{\text{NP}}\ \textit{dog}]_2] \rrbracket(\llbracket \textit{John}_1 \rrbracket)$
$= \lambda x \lambda ghi{:}\ _{g_1=\text{F}(J),\ g_2=\text{F}(\textit{dog}),\ \text{F}(\text{C})(g_2)}\ [g <_3 h,\ h_2(ih_3),\ \text{F}(\textit{own})(ih_3h_1)]$

Notice that the indexed head noun *dog*$_2$, cf. (40d), is interpreted as a dynamic predicate of type **eaast** that presupposes for its input assignment that the dref 2 is anchored to the property 'dog', and identifies this dref with the argument x. The condition F(C)(P) expresses that P is a count property; this implements the treatment of the count/mass distinction as an agreement feature, as discussed in Section 2. The indexed determiner *a*$_3$, cf. (40c), takes this meaning as an argument, introduces a dref 3 that falls under the property of dref 2, and creates a meaning that combines with the verbal predicate, cf. (40e). The input assignment g of (40g) must contain the drefs 1 and 2 anchored to John and to the property 'dog'. The latter condition again just requires that the dref is anchored to a meaning, which can be accommodated uncontroversially when (40) updates a context c, as in (41). The output h differs from the input g insofar as it also contains a dref 3 that is anchored to a dog that John owns.

(41) $c_0 + (40)$
$= \{\langle i,h \rangle \mid \exists g{:}\ _{g_1=\text{F}(J),\ g_2=\text{F}(\textit{dog}),\ \text{F}(\text{C})(g_2)}\ [\langle i,g \rangle \in c_0,\ g <_3 h,\ h_2(i)(h_3),\ \text{F}(\textit{own})(ih_3h_1)],\ = c_1$

We consider first the continuation with an object-related anaphor, (42), where I assume that pronouns come with presuppositions (here, *it* with the feature non-human). The update of c_1 with (42) is given in (43) in a somewhat simplified manner, with presupposed material in smaller face. Notice that the presupposition of *it*$_3$, that the input assignment contains the dref 3 and that it is anchored to a non-human entity is satisfied at the point where it is interpreted if we restrict the models of interpretations to those that satisfy $\forall i \forall x[F(i)(\text{DOG}) \rightarrow F(i)(\text{NONHUM})]$.

(42) a. $[\![it_3]\!] = \lambda P \lambda ghi \ {\scriptstyle F(\text{NONHUM})(ig_3)} \ [P(g_2)(ghi)]$
b. $[\![\textit{is friendly}]\!] = \lambda x \lambda ghi[g{=}h, F(\textit{friendly})(ix)]$
c. $[\![\textit{it}_3 \textit{ is friendly}]\!] = \lambda ghi \ {\scriptstyle F(\text{NONHUM})(ig_3)} \ [g{=}h, F(\textit{friendly})(ig_3)]$

(43) $c_1 + (42) = \{\langle i,h\rangle \mid \exists g{:}\ {\scriptstyle g_1=F(\textit{John}),\ g_2=F(i)(\text{dog}),\ F(C)(g_2)}\ [\langle i,g\rangle \in c_0,$
$g{<_3}h,\ h_2(ih_3),\ F(\textit{own})(ih_3h_1),\ {\scriptstyle F(\text{NONHUM})(ih_3)},\ F(\textit{friendly})(ih_3)]\}$

Let us now consider the influence of negation, which we interpret here as a VP operator:

(44) a. $[\![\textit{John}_1]\!] = \lambda P \lambda ghi \ {\scriptstyle g_1=F(\textit{John})} \ [P(g_1)(ghi)]$
b. $[\![\textit{doesn't}]\!] = \lambda P \lambda x \lambda ghi \ [g{=}h, \neg\exists k[P(g)(k)(i)(x)]]$
c. $[\![\textit{own} \ [_{\text{DP}}\ a_3\ [_{\text{NP}}\ \textit{dog}]_2]]\!]$
$= \lambda x \lambda ghi{:}\ {\scriptstyle g_2=F(\textit{dog}),\ F(C)(g_2)}\ [g{<_3}h,\ n(h_3){=}1,\ h_2(ih_3),\ F(\textit{own})(ih_3x)]$
d. $[\![\textit{doesn't own} \ [_{\text{DP}}\ a_3\ [_{\text{NP}}\ \textit{dog}]_2]]\!] = [\![\textit{doesn't}]\!]([\![\textit{own} \ [_{\text{DP}}\ a_3\ [_{\text{NP}}\ \textit{dog}]_2]]\!])$
$= \lambda x \lambda ghi{:}\ {\scriptstyle g_2=F(\textit{dog}),\ F(C)(g_2)}\ [g{=}h, \neg\exists k[g{<_3}k,\ k_2(ik_3),\ F(\textit{own})(ik_3x)]]$
e. $[\![\textit{John}_1 \textit{ doesn't own} \ [_{\text{DP}}\ a_3\ [_{\text{NP}}\ \textit{dog}]_2]]\!]$
$= \lambda x \lambda ghi{:}\ {\scriptstyle g_1=F(J),\ g_2=F(i)(\textit{dog}),\ F(C)(g_2)}\ [g{=}h, \neg\exists k[g{<_3}k,\ k_2(ik_3),\ F(\textit{own})(ik_3k_1)]]$

(45) $c_0 + (44)$
$= \{\langle i,g\rangle \mid \exists g{:}\ {\scriptstyle g_1=F(J),\ g_2=F(\textit{dog}),\ F(C)(g_2)}\ \langle g,i\rangle \in c_0, \neg\exists k[g{<_3}k,\ k_2(ik_3),\ F(\textit{own})(ik_3k_1)]]\} = c_1$

Update of c_0 with (44) does not introduce the dref 3 in the output assignment as it is in the scope of the negation. However, the presupposed property of dref 2 is accessible, and can license concept anaphora or kind anaphora. Example (46) illustrates this with concept anaphora.

(46) a. $[\![\textit{Mary}_4]\!] = \lambda P \lambda ghi{:}\ {\scriptstyle g_4=F(M)}\ [P(g_4)(ghi)]$
b. $[\![\textit{own}]\!] = \lambda y \lambda x \lambda ghi[g{=}h, F(\textit{own})(iyx)]$
c. $[\![\textit{one}_5]\!] = \lambda P \lambda R \lambda x \lambda ghi \exists k \exists k'[g{<_5}k,\ \#(k_5){=}1,\ P(k_5)(kk'i),\ R(k'_5x)(k'hi)]$
d. $[\![[_{\text{NP}}\ ]_2]\!] = \lambda x \lambda ghi\ [g{\leq}h \wedge g_2{=}x]$
e. $[\![[\textit{one}_5\ [_{\text{NP}}\ ]_2]]\!] = [\![\textit{one}_5]\!]([\![[_{\text{NP}}\ ]_2]\!])$
$= \lambda R \lambda x \lambda ghi \exists k \exists k'[g{<_5}k,\ \#(k_5){=}1,\ k{\leq}k',\ g_2(k_5)(kk'i),\ R(k'_5x)(k'hi)]$
f. $[\![\textit{own}\ [\textit{one}_5\ [_{\text{NP}}\ ]_3]]\!] = [\![[\textit{one}_5\ [_{\text{NP}}\ ]_2]]\!]([\![\textit{own}]\!])$
$= \lambda x \lambda ghi \exists k[g{<_5}k,\ \#(k_5){=}1,\ g_2(k_5)(khi),\ F(\textit{own})(ik_5x)]$
h. $[\![\ \textit{Mary}_4\ \textit{own}\ [\textit{one}_5\ [_{\text{NP}}\ ]_2]]\!]$
$= \lambda ghi{:}\ {\scriptstyle g_4=F(M)}\ \exists k[g{<_5}k,\ \#(k_5){=}1,\ g_2(k_5)(khi),\ F(\textit{own})(ik_5g_4)]$

According to (46d), the anaphoric empty NP presupposes that the dref 2 exists in the input assignment g, and is anchored to a meaning of type **eaast**. The meaning distinguishes between input g and output h, allowing for concepts that introduce their own dref, as in *John doesn't own a dog from a shelter. Mary owns one.* The determiner *one* introduces a dref 6 anchored to an object that belongs to the concept of the dref 2.

When (46h) updates c_1 of (45), we arrive at the representation (47). Notice that the condition $g_2(h_5)(ghi)$ is to be spelled out as $F(\textit{dog})(ih_2)$, but the reference to this property is mediated by the concept dref 2.

(47) $c_1 + (46) = \{\langle i,h\rangle \mid \exists g\text{: } g_1{=}F(J), g_2{=}F(\mathit{dog}), F(\textsc{c})(g_2), g_4{=}F(M), [\langle i,g\rangle \in c_1,$
$\neg\exists k[g{<_3}k, \#(k_3){=}1, k_2(ik_3), F(\mathit{own})(ik_3k_1)]], g{<_5}h, n(h_5){=}1, g_2(ih_5), F(\mathit{own})(ik_5g_4)]]\}$

As for kind anaphora, consider the following continuation of the output context c_1 of (45). The kind pronoun *them*$_{24}$ picks up the concept dref 2, provided that this dref is anchored to a count concept, following the discussion of kind anaphora picking up mass vs. count antecedents in (18). It introduces a dref 4 that refers to the individual concept that maps indices to the sum of the pluralities that fall under the anchor of dref 2. As in the case of concept anaphors in (46)/(47), this is possible as the dref 2 is accessible, in spite of the negation in the first clause. Notice that the kind-referring pronoun *it* would not be possible as it expects that the anchor of its dref has the mass property, and we should assume mutual exclusivity of the count and mass properties, $\forall P \neg[F(\textsc{m})(P) \wedge F(\textsc{c})(P)]$.

(48) a. $[\![\mathit{he}_1]\!] = \lambda P\lambda ghi\text{: } F(\textsc{male})(g_1)\ [g{=}h, P(g_1)(ghi)]$
b. $[\![\mathit{hate}]\!] = \lambda y\lambda x\lambda ghi[g{=}h, F(\mathit{hate})(iyx)]$
c. $[\![\mathit{them}_{24}]\!] = \lambda R\lambda x\lambda ghi\text{: } F(\textsc{c})(g_2)\ \exists k[g{<_4}k \wedge k_4{=}\lambda i^{\cap\sqcup}g_2(i), R(k_4x)(khi)]$
d. $[\![\mathit{hate\ them}_{24}]\!] = [\![\mathit{them}_{24}]\!]([\![\mathit{hate}]\!])$
$= \lambda x\lambda ghi\text{: } F(\textsc{c})(g_2)\ [g{<_4}k, h_4{=}\lambda i^{\cap\sqcup}g_2(i), F(\mathit{hate})(ih_4x)]$
e. $[\![\mathit{he}_1\ \mathit{hate\ them}_{24}]\!] = [\![\mathit{he}_1]\!]([\![\mathit{hate\ them}_{24}]\!])$
$= \lambda ghi\text{: } F(\textsc{c})(g_2), F(\textsc{male})(g_1), [g{<_4}k, h_4{=}\lambda i^{\cap\sqcup}g_2(i), F(\mathit{hate})(ih_4h_1)]$

(49) $c_1 + (48)$
$= \{\langle i,h\rangle \mid \exists g\text{: } g_1{=}F(J), g_2{=}F(\mathit{dog}), F(\textsc{c})(g_2), F(\textsc{male})(g_1)\ [\langle i,g\rangle \in c_1,$
$\neg\exists k[g{<_3}k, \#(k_3){=}1, k_2(ik_3), F(\mathit{own})(ik_3k_1)]], g{<_4}h, h_4{=}\lambda i^{\cap\sqcup}g_2(i), F(\mathit{hate})(ih_4h_1)]]\}$

Let us now compare concept anaphors like in (46) with partitive anaphors as in (50):

(50) a. $[\![\textsc{past}_1\ \mathit{John}_2\ \mathit{buy}\ [_{\textsc{dp}}\ \mathit{two}_4\ [\mathit{apples}]_3]]\!]$
$= \lambda ghi\text{: } g_2{=}F(J), g_3{=}\lambda i\ {}^{\sqcup}F(\mathit{apple})(i)\ [g{<_{14}}k, g_1{<}i, g_3(g_1g_4), \#(g_4){=}2, F(\mathit{buy})(g_1g_4g_2)]$
b. $[\![\textsc{past}_5\ \mathit{Mary}_6\ \mathit{eat}\ [_{\textsc{dp}}\ \mathit{one}_7\ [_{\textsc{pp}}\ ]_4]]\!]$
$= \lambda ghi\text{: } g_6{=}F(M)\ [g{<_{57}}h, g_5{<}i, g_7{\sqsubseteq}g_4, \#(g_7){=}1, F(\mathit{eat})(g_5g_7g_6)]$

(51) $c_0 + (50a) + (50b)$
$= \{\langle i,h\rangle \mid \exists g\text{: } g_2{=}F(J), g_3{=}\lambda i^{\sqcup}F(\mathit{apple})(i), g_6{=}F(M)\ [g{<_{1457}}h, g_1{<}i, g_5{<}i,$
$g_3(g_1g_4), \#(g_4){=}2, F(\mathit{buy})(g_1g_4g_2), g_7{\sqsubseteq}g_4, \#(g_7){=}1, F(\mathit{eat})(g_5g_7g_6)]$

The partitive interpretation has to be spelled out as *one of them*, not as *one apple*. As we have seen, this is because nominal concept anaphors and partitive anaphors differ in their syntactic structure; one involves an empty NP, the other, an empty PP. Both are anaphoric, but while NP anaphors pick up a dref anchored to a property, e.g. $[\![[_{\textsc{np}}\ ]_3]\!] = \lambda x\lambda ghi\text{: } g_3{=}x\ [g{=}h]$, partitive PP anaphors pick up a dref anchored to an individual, e.g. $[\![[_{\textsc{pp}}\ ]_5]\!] = \lambda x\lambda ghi\text{: } x{\sqsubseteq}g_5\ [g{=}h]$.

There are other types of anaphors that can be treated as concept anaphor, like VP ellipsis (cf. Asher 1993, Hardt 1999). Example (52) illustrates this with the sentence *Mary doesn't eat an apple. John does.* The VP in (52a) gets indexed via an identity presupposition in (52b), which stays accessible under negation in (52c). In the second sentence, the empty VP can anaphorically pick up that dref, cf. (52e).

(52) a. $[\![[_{\textsc{vp}}\ \mathit{eat}\ [_{\textsc{dp}}\ \mathit{an}_5\ [_{\textsc{np}}\ \mathit{apple}]_4]]]\!] = \lambda x\lambda ghi\text{:} g_4{=}F(\mathit{apple}), F(\textsc{c})(g_4)\ [g{<_5}h, g_4(ih_5), F(\mathit{eat})(ih_5x)]$
b. $[\![[_{\textsc{vp}}\ \mathit{eat}\ [_{\textsc{dp}}\ \mathit{an}_5\ [_{\textsc{np}}\ \mathit{apple}]_4]]_6]\!]$

= $\lambda x\lambda ghi$: g_4=F(*apple*), F(C)(g_4), g_6=$\lambda x\lambda ghi$ [$g<_5h$, $g_4(ih_5)$, F(*eat*)(ih_5x)] [$g\leq h$, $g_6(x)(ghi)$]

c. ⟦[*doesn't eat* [DP *an*5 [NP *apple*]4]]6]⟧
= $\lambda x\lambda g\lambda h\lambda i$: g_4=F(*apple*), F(C)(g_4), g_6=$\lambda x\lambda ghi$[$g<_5h$, $g_4(ih_5)$, F(*eat*)(ih_5x)]
[$g=h$, $\neg\exists k[g\leq k$, $g_6(x)(gki)$]

d. ⟦[*Mary*1 [*doesn't eat* [DP *an*5 [NP *apple*]4]]6]⟧
= $\lambda x\lambda g\lambda h\lambda i$: g_1=F(*M*), g_4=F(*apple*), F(C)(g_4), g_6=$\lambda x\lambda ghi$[$g<_5h$, $g_4(ig_5)$, F(*eat*)(ih_5x)]
[$g=h$, $\neg\exists k[g\leq k$, $g_6(h_1)(gki)$]]

e. ⟦[*John*2 [*does* [VP]6]]]⟧
= λghi: g_2=F(*M*), g_4=F(*apple*), F(C)(g_4), g_6=$\lambda x\lambda ghi$[$g<_5h$, $g_4(ig_5)$, F(*eat*)(ih_5x)] [$g\leq h$, $g_6(g_2)(ghi)$]

Notice that the dref 6 is anchored to a semantic object that is itself dynamic, of type **eaast**, and that the second sentence introduces a dref 5 for an apple that John eats. Dynamic bindings are also required for *one*-anaphora when the antecedent introduces drefs, as in *Mary saw a dog with a flea collar and John saw one too.*

It should be noted that concept anaphora work across turns, as in (53). They also occur as exophoric expressions provided the context provides an appropriate entity, where *one* appears slightly degraded as first item, cf. (54a,b).

(53) A: *John doesn't own **a dog**.*
B: *I know. He is afraid of **them**. But his wife Mary would like to have **one**.*

(54) [On seeing a tarantula in a terrarium at a house party]:
a. *I would not like to have* ?***one**. I am afraid of **them**.*
b. *I am afraid of **them**. I would not like to have **one**.*

This is compatible with our representation if contexts c are understood as common grounds, that is, as information states that are supposed to be shared by the participants. Example (54) shows that concept and kind expressions can be build on non-linguistic, purely situational antecedents, which requires a representation of the shared situational knowledge (cf. Buch 2023).

5 Conclusion

The topic of this article were two related types of anaphora, called concept and kind anaphora, as in *Mary owns a dog. John owns **one** too. He likes **them***. The main points were the following:

First, in order to treat the differences between count nouns and mass nouns for kind anaphora in terms of number agreement (as in *Mary bought two packages of rice. She likes it.* vs. *Mary bought a package of beans. She likes them.*), I proposed morphosyntactic features MASS vs. COUNT (see Krifka to appear for a treatment as an ontological distinction). Second, I proposed that DPs with a head noun like *a dog* introduce a discourse referent for a concept, here the sense of the head noun *dog*, which can be picked up by empty NPs as in [DP *one* [NP]], or *one*-NPs as in [DP *a* [NP *small* [NP *one*]]]. Such concepts can also sponsor kind discourse referents. This is similar to an earlier proposal by LuperFoy (1991). Third, in order to treat the insensitivity against anaphoric islands like negation, as in *Mary does not own a dog. But John owns one. He likes them*, I proposed that concept discourse referents escape anaphoric islands, similar to names. This can be explained as concept discourse referents are identified with the concept, just as a name is identified with the carrier of the name. I developed a fully compositional account within the framework of dynamic semantics. I also showed how concept anaphora differ from partitive anaphora. While both can be expressed by empty constituents, the former involve empty NPs, and the latter, PPs, as in *Mary bought apples. Max bought* [$_{DP}$ *one* [$_{NP}$]] *too.* vs *Mary bought apples. Max ate* [$_{DP}$ *one* [$_{PP}$]] (in the reading: *one of them*). This opens the door to explain other cases of ellipsis as involving anaphora, e.g. VP ellipsis.

There are several points that should be developed further. The first concerns the theory of the mass/count distinction, which should be tested with additional data, and compared to other theories. The second is our observation that not every concept can sponsor kind anaphora. We have seen that kinds should not depend on features of the situational context (as in *dogs in this kennel*), there might be further restrictions. There is also is the distinction between kinds in general and well-established kinds that can be referred to by singular definite generic terms. Another point is the choice of discourse referents. We have assumed that syntactic representations come with indices that determine the choice of drefs. This is not only a simplification but also leads to wrong results. For example, the treatment of VP ellipsis in (52) runs into problems with cases like *Mary* [$_{VP}$ *ate* [*an*$_2$ *apple*]]$_1$ *and John did* 1 *too*, as the first clause introduces an object discourse referent 2 anchored to the apple that Mary ate, which precludes that the second sentence is interpreted, as the condition $g{<_2}h$ presupposes that the input assignment g does not contain the index 2 in its domain. What we need is a representation for indefinites with conditions that introduce, given an input assignment g, the “next available” discourse referent. In the same way, anaphoric expressions would not get their discourse referents from syntactic indexing but from the syntactic category, the semantic type, the conceptual plausibility, and recency phenomena as described in centering theory (cf. e.g. Hardt 1999). Finally, the anaphors discussed here require generally more complex semantic representations, which should result in slower processing (cf. Gernsbacher 1991) and differences in salience (cf. Modarresi & Krifka 2021, Krifka & Modarresi 2023 for anaphors to incorporated nominals and weak definites).

References

Asher, N. 1993. *Reference to abstract objects in discourse*. Dordrecht: Kluwer.

Bale, Alan. 2021. Ontology, number agreement, and the count-mass distinction. Ed. Tibor Kiss, Francis Jeffry Pelletier, & Hailima Husić, *Things and stuff. The semantics of the count-mass distinction*. Cambridge: Cambridge University Press. 237-260.

Buch, F. 2024. *Speech act deixis. A situated dynamic account for observational and experimental insights in spoken German.* Doctoral dissertation, Humboldt-Universität zu Berlin. https://doi.org/10.18452/28552

Carlson, G. N. 1977. *Reference to kinds in English.* Doctoral dissertation, University of Massachusetts.

Chierchia, G. 1998. Reference to kinds across languages. *Natural Language Semantics* 6, 339-405.

Cooper, R. 2012. Type theory and semantics in flux. Ed. Ruth M. Kempson, Tim Fernando, & Nicholas Asher, *Philosophy of Linguistics, Handbook of the Philosophy of Science*. Elsevier. 271-323.

Dahl, D. 1984. *The structure and function of ‘One’-anaphora in English*. Doct. Diss., U Minnesota.

Gernsbacher, M. 1991. Comprehending conceptual anaphors. *Language and Cognitive Processes* 6, 81-105.

Grimm, S. 2012. Degrees of countability: A mereotopological approach to the mass/count distinction. *SALT* 22, 1-15.

Hardt, D. 1999. Dynamic interpretation of VP ellipsis. *Linguistics & Philosophy* 22, 185-219.

Heim, I. 1983. File change semantics and the familiarity theory of definiteness. Ed. R. Bäuerle & et al., *Meaning, Use and the Interpretation of Language*. Berlin: Walter de Gruyter. 164-190.

Hofmann, L. 2022. *Anaphora and negation.* Doct. dissertation, U of California at Santa Cruz.

Jackendoff, R. 1977. *X-bar Syntax: A Study of Phrase Structure*. Cambridge MA: MIT Press.

Kamp, H, & Reyle, U. 1993. *From discourse to logic. Introduction to model theoretic semantics of natural language, formal logic, and Discourse Representation Theory*. Dordrecht: Kluwer.

Karttunen, L. 1969. Discourse referents. *Coling 1969*. Also in J. D. McCawley 1976, Syntax and Semantics 7: *Notes from the linguistic underground.* New York: Academic Press, 363-385.

Krifka, M. 1993. Focus and presupposition in dynamic interpretation. *Journal of Semantics* 10, 269-300.

Krifka, M. 2021a. Individuating matter over time. Ed. Hana Filip, *Countability in natural language*. Cambridge University Press. 121-144.

Krifka, M. 2021b. Counting possible configurations. Ed. Sebastian Löbner et al., *Concepts, Frames and Cascades in Semantics, Cognition and Ontology. Language, Cognition, and Mind*. Cham: Springer. 43-63.

Krifka, M. To appear. Kind anaphora and the mass/count distinction.

Krifka, M. et al. 1995. Genericity: an introduction. Ed. G N. Carlson & F. J. Pelletier, *The generic book*. University of Chicago Press. 1-124.

Krifka, M, & Modarresi, F. 2023. Anaphoric potential of bare nominals, incorporated objects and weak definites in German. Ed. L Gardelle et al., *Studies in Language Companion Series: Reference*. Amsterdam: John Benjamins, 27-51.

Lewis, D. 1979. Scorekeeping in a language game. *Journal of Philosophical Logic* 8, 339-359.

Lødrup, H. 2012. Some Norwegian “type anaphora” are surface anaphora. *Journal of Germanic Linguistics* 24, 23-52.

LuperFoy, S. 1991. Discourse Pegs: A computational analysis of context-dependent referring expressions. *Doctoral Dissertation, University of Texas at Austin*.

Miller, A. & Pullum, G. K. 2014. Exophoric VP ellipsis. Ed. P. Hofmeister & E. Norcliffe, *The core and the periphery: Data-driven perspectives on syntax inspired by Ivan A. Sag*. Chicago: University of Chicago Press.

Modarresi, F. & M. Krifka. 2021. Pseudo Incorporation and Anaphoricity: Evidence from Persian. *Glossa* 6.

Moltmann, F. 1998. Part structures, integrity, and the mass-count distinction. *Synthese* 116, 75-111.

Muskens, R. 1996. Combining Montague semantics and discourse representation. *Linguistics and Philosophy* 19, 143-186.

Recasens, M et al. 2016. Sense anaphoric pronouns: Am I One? *Workshop on Coreference Resolution*. Association of Computational Linguistics.

Roberts, C. 1989. Modal subordination and pronominal anaphora in discourse. *Linguistics and Philosophy* 12, 683-721.

Rooth, M. 1987. Noun phrase interpretation in Montague Grammar, File Change Semantics, and Situation Semantics. Ed. P. Gärdenfors, *Generalized quantifiers. Linguistic and logical approaches*. Dordrecht: Reidel. 237-268.

Snider, T. N. 2017. *Anaphoric reference to propositions.* Doct. Diss., Cornell University.

Sutton, P. R., & H. Filip. 2016. Mass/count variation: A mereological, two-dimensional semantics. *The Baltic International Yearbook of Cognition, Logic and Communication* 11, 1-45.

Webber, B. L. 1979. *A formal approach to discourse anaphora*. New York: Garland.

Weeber, F. 2023. *The German indefinite pronoun* ein(er). *Interpretation and discourse linking.* Doctoral dissertation, Universität zu Köln.